

The gamma-composition kernel is strictly TP_2 , but not TP_3

Run `fa_banach_001`, agent lane 10

2026-08-11

Source and open problems

For $s, x > 0$, write

$$Q(s, x) = \frac{\Gamma(s, x)}{\Gamma(s)} = \frac{1}{\Gamma(s)} \int_x^\infty t^{s-1} e^{-t} dt$$

for the regularized upper incomplete gamma function, and let $Q^{-1}(s, v)$ denote its inverse in the second variable. Richards and Uhler [1] introduced, for fixed $c > 0$,

$$\mathcal{C}_c(u, v) = Q(u + c, Q^{-1}(u, v)), \quad u > 0, \quad 0 \leq v < 1.$$

In Remark 4.4 they report substantial numerical evidence that $(u, v) \mapsto \mathcal{C}_c(u, v)$ is both STP_2 and STP_3 , and pose these as open problems. Their decreasing-order convention asks for positive minors when $u_1 > \dots > u_r$ and $v_1 > \dots > v_r$.

Remark 4.4. As we noted before, [Furman and Zitakis \(2008c\)](#) proved that the function $u \mapsto \mathcal{C}_c(u, v)$ is decreasing, a result which raises the issue of whether the function $(c, u) \mapsto \mathcal{C}_c(u, v)$, $c > 0$, $u > 0$ is RR_2 .

Let F_u denote the cumulative distribution function of a gamma-distributed random variable having index parameter u , i.e., a random variable whose probability density function is the integrand in (4.5). Then, $\mathcal{C}_c(u, v) = 1 - F_{c+u}(Q^{-1}(u, v))$. Since $Q(u, v) = 1 - F_u(v)$ then $Q^{-1}(u, v) = F_u^{-1}(1 - v)$, so we obtain

$$\mathcal{C}_c(u, v) = 1 - F_{c+u}(F_u^{-1}(1 - v)).$$

We performed extensive calculations using this identity and determined that the function $(c, u) \mapsto \mathcal{C}_c(u, v)$ is not RR_2 as many of its 2×2 determinants are positive. Further, this function is not TP_2 , although it seems to fail barely to be so; for $v = 0.211$, $(c_1, c_2) = (4.047, 1.210)$, and $(u_1, u_2) = (3.203, 0.189)$, we obtained $\det(\mathcal{C}_{c_i}(u_j, v)) = -0.026$, and this negative value of the determinant was similar in magnitude to all other negative values that we found.

We also carried out calculations regarding the total positivity properties of the function $(u, v) \mapsto \mathcal{C}_c(u, v)$ and found substantial evidence that this function is both STP_2 and STP_3 . As we have not been able to establish such results analytically, we pose them as open problems.

Answer and proof intuition

The two orders behave differently. Differentiate $\mathcal{C}_c$ in v . The Jacobian of the inverse gamma CDF cancels the old gamma density, leaving a positive factor times the c -th power of the relevant

gamma quantile. Saunders and Moran's strict quantile-ratio theorem [2] says exactly that these powers form an STP_2 kernel. Because all rows of $\mathcal{C}_c$ start at the common value zero when $v = 0$, integration preserves strict order-two positivity.

Order three fails. A simple rational parameter choice gives a determinant near -1.21×10^{-4} . We certify its sign by positive-series enclosures of the incomplete gamma ratios, monotone bracketing of all inverse quantiles, and outward-rounded interval arithmetic.

Theorem 1 (Resolution of the two questions). *For every $c > 0$, the kernel $\mathcal{C}_c$ is STP_2 on $(0, \infty) \times (0, 1)$. It extends to a TP_2 kernel on $(0, \infty) \times [0, 1]$, but the extension is not strict at $v = 0$.*

The proposed general STP_3 assertion is false. Indeed, at $c = 1/8$, the decreasing triples

$$(u_1, u_2, u_3) = \left(100, \frac{18}{5}, \frac{6}{25}\right), \quad (v_1, v_2, v_3) = \left(\frac{99}{100}, \frac{63}{100}, \frac{3}{20}\right)$$

give

$$\det(\mathcal{C}_{1/8}(u_i, v_j))_{i,j=1}^3 < 0.$$

Thus the two open problems have answers yes at order two and no at order three.

The positive STP_2 theorem

Let F_u be the CDF and $f_u(x) = x^{u-1}e^{-x}/\Gamma(u)$ the density of a unit-scale gamma variable of shape u . Put

$$q_u(p) = F_u^{-1}(p), \quad x_u(v) = q_u(1 - v).$$

Then

$$\mathcal{C}_c(u, v) = 1 - F_{u+c}(x_u(v)).$$

Implicit differentiation of $F_u(x_u(v)) = 1 - v$ gives

$$\frac{\partial x_u}{\partial v} = -\frac{1}{f_u(x_u(v))}.$$

Consequently

$$h_c(u, v) := \frac{\partial}{\partial v} \mathcal{C}_c(u, v) = \frac{f_{u+c}(x_u(v))}{f_u(x_u(v))} = \frac{\Gamma(u)}{\Gamma(u+c)} x_u(v)^c. \quad (1)$$

Saunders and Moran proved that, whenever $0 < \beta < \alpha < 1$,

$$u \mapsto \frac{q_u(\alpha)}{q_u(\beta)} \quad \text{is strictly decreasing on } (0, \infty). \quad (2)$$

Fix $u_1 > u_2 > 0$ and $v_1 > v_2$. Applying (2) with $\beta = 1 - v_1$ and $\alpha = 1 - v_2$, and rearranging, yields

$$\frac{x_{u_1}(v_1)}{x_{u_2}(v_1)} > \frac{x_{u_1}(v_2)}{x_{u_2}(v_2)}. \quad (3)$$

By (1), the ratio

$$\frac{h_c(u_1, v)}{h_c(u_2, v)} = \frac{\Gamma(u_1)\Gamma(u_2+c)}{\Gamma(u_1+c)\Gamma(u_2)} \left(\frac{x_{u_1}(v)}{x_{u_2}(v)}\right)^c$$

is therefore strictly increasing in v . Equivalently, h_c is STP_2 .

For completeness, strict positivity survives integration here by an elementary identity. Let $h_1, h_2 > 0$, suppose $r = h_1/h_2$ is strictly increasing, and set $H_i(v) = \int_0^v h_i(t) dt$. Then

$$\left(\frac{H_1}{H_2}\right)'(v) = \frac{h_2(v)}{H_2(v)^2} \int_0^v (r(v) - r(t)) h_2(t) dt > 0. \quad (4)$$

Since $\mathcal{C}_c(u, 0) = 0$, equation (1) gives $\mathcal{C}_c(u, v) = \int_0^v h_c(u, t) dt$. Taking $h_i(t) = h_c(u_i, t)$ in (4) proves that $\mathcal{C}_c(u_1, v)/\mathcal{C}_c(u_2, v)$ is strictly increasing. This is precisely the positivity of every ordered 2×2 minor, proving STP_2 on the interior.

If one of the ordered v 's equals zero, the corresponding column is zero. The determinant is then nonnegative but not positive. This proves the exact boundary statement in Theorem 1.

The certified 3×3 counterexample

At the parameters in Theorem 1, the kernel matrix is, to 18 decimal places,

$$\begin{pmatrix} 0.990341980391842966 & 0.634744085941809884 & 0.152888248142902939 \\ 0.992092691738034760 & 0.656560587655528763 & 0.165231530309850930 \\ 0.999117517495731797 & 0.785807279598486534 & 0.233184274842588704 \end{pmatrix}.$$

Its high-precision determinant is approximately

$$-0.0001210585212763807470565673499421878.$$

The following certification removes any reliance on rounded displayed values. For $s, x > 0$, use the positive series

$$P(s, x) := F_s(x) = \frac{x^s e^{-x}}{\Gamma(s+1)} \sum_{n=0}^{\infty} \frac{x^n}{(s+1)_n}. \quad (5)$$

After the ratio of successive terms becomes less than $1/2$, the remaining tail is at most twice the next term. The accompanying script evaluates (5) with outward-rounded intervals. For each pair (u, v) , it first finds numbers $x_- < x_+$ and certifies

$$P(u, x_-) < 1 - v < P(u, x_+).$$

Monotonicity then encloses the kernel entry by

$$1 - P(u + c, x_+) \leq \mathcal{C}_c(u, v) \leq 1 - P(u + c, x_-).$$

Set $D = \det(\mathcal{C}_{1/8}(u_i, v_j))$. Interval evaluation of its six-term expansion gives the outward-rounded enclosure

$$\begin{aligned} -1.210585212763807470565673499423 \times 10^{-4} &\leq D, \\ D &\leq -1.210585212763807470565673499421 \times 10^{-4}. \end{aligned}$$

The upper endpoint is strictly negative, proving failure of TP_3 , hence of STP_3 .

Verification, novelty, and scope

The interval certificate is in `code/interval_certificate.py` and uses 80 decimal digits of outward-rounded interval arithmetic. Its approximate root finder is not trusted: every root is independently bracketed by the certified series before it enters the determinant calculation.

On 2026-08-11, exact-title, exact-kernel, STP terminology, and citation searches found the source and its published version but no later paper claiming a resolution of these two questions. The run's four lightweight indexes and local arXiv corpus likewise contained no matching result. This is a bounded novelty check, not an exhaustive review of all non-arXiv literature.

The theorem proves STP_2 for every $c > 0$ and refutes the proposed general STP_3 statement with one certified c . It does not classify those individual values of c , if any, for which the order-three kernel might still be strictly totally positive. Additional searches beyond the certified small- c witness produced only cancellation-scale signals for larger c , so no such classification is asserted.

Human review recommendation

Review as a likely-valid full resolution. The two highest-value independent checks are: (i) verify the direction change from (2) to (3); and (ii) run the interval certificate and confirm that its determinant upper endpoint is negative.

References

- [1] Donald St. P. Richards and Caroline Uhler, *Loading Monotonicity of Weighted Premiums, and Total Positivity Properties of Weight Functions*, arXiv:1806.07957 (2018).
- [2] I. W. Saunders and P. A. P. Moran, *On the quantiles of the gamma and F distributions*, Journal of Applied Probability **15** (1978), no. 2, 426–432, doi:10.2307/3213414.